

Planning as Autoregressive Sequence Modelling: Decoding Strategies

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Introduction

Planning is a core problem in artificial intelligence: given a model of an environment and a desired objective, an agent must construct a sequence of actions that leads from an initial state to a goal state while satisfying constraints and optimising some notion of cost or reward. Classical approaches formulate planning as search over a state space (e.g., A*) [1], whereas reinforcement learning (RL) frames it as maximising cumulative reward through value functions or policies [2]. More recently, a generative modelling perspective has emerged, in which trajectories themselves are treated as structured sequences that can be modelled probabilistically. From this viewpoint, planning becomes a problem of inference in a learned distribution over trajectories rather than explicit search in a hand-designed state graph [3].

In this project, planning in a 2D gridworld is formulated as autoregressive sequence modelling. A trajectory is represented as an ordered sequence of states (or state-action pairs), and a model is trained using next-step prediction on shortest paths generated by A* [1]. Once trained, the model defines a learned probability distribution over feasible trajectories. Planning then corresponds to decoding: generating a trajectory from this distribution that satisfies the task objective. In this framing it is useful to distinguish between autoregressive generation and decoding. Autoregressive generation means that the model produces a trajectory sequentially, predicting each next state (or state-action token) conditioned on the previously generated tokens. Decoding refers to the rule used during inference to convert the model's predicted probability distribution into an actual trajectory. Decoding is necessary because multiple continuations are typically possible at each

step, and a procedure is required to decide which one forms the final plan [3]. More precisely, The central question of this project is how different decoding strategies affect the quality, reliability, diversity, and computational cost of the generated plans. While the underlying model is fixed after training, inference procedures such as greedy decoding, stochastic sampling with temperature control, and beam search may extract substantially different trajectories from the same distribution [4]. Greedy decoding prioritises local likelihood, potentially sacrificing global optimality or diversity. Stochastic sampling can increase diversity but may introduce instability or suboptimal paths. Beam search balances exploration and exploitation at the cost of increased computation [4]. By systematically comparing these strategies, the project investigates how inference choices shape planning outcomes when planning is cast as generative sequence modelling.

Background

Generative Models

Generative models are types of machine learning systems that aim to represent a probability distribution over data, or to draw samples from that distribution [5, p. 493]. Specifically, if $\mathbf{x}$ is a collection of features observed from an object or event, like words in a sentence or coordinates in a maze, and $p(\mathbf{x})$ is the true joint distribution of those features, a generative model learns an approximation $\hat{p}(\mathbf{x})$ of this distribution from training observations [5]. Once such a distribution has been learned, it can be used to generate new observations that are similar to the observations in the training data by sampling from $\hat{p}(\mathbf{x})$ [5, p. 102].

The Planning Problem

Broadly, automated planning is the field of research addressing the problem of finding an ordered or partially ordered series of actions that changes an environment from its initial state to a desired goal state [6].

In this work, the planning problem is to generate a valid trajectory from a given starting position to a specified goal position in a 2D Maze environment. In this environment, a trajectory refers to a sequence of maze positions connecting a start location to a goal location without passing through walls.

Assuming a discrete grid-based representation, maze coordinates are integers identifying grid cells. A trajectory can therefore be represented as a sequence of coordinates in which each consecutive pair corresponds to adjacent non-wall cells, with adjacency restricted to horizontal and vertical moves.

insert visualisation of the maze environment here.

References

- [1] P. E. Hart, N. J. Nilsson, and B. Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.
- [2] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.
- [3] Michael Janner, Qiyang Li, and Sergey Levine. Offline Reinforcement Learning as One Big Sequence Modeling Problem. *Advances in Neural Information Processing Systems*, 2021.
- [4] Daniel Jurafsky and James H. Martin.
- [5] Ian Goodfellow, Yoshua Bengio, Aaron Courville. Deep Learning.
- [6] Diaeddin Alarnaouti, George Baryannis, Mauro Vallati. Reformulation techniques for automated planning: a systematic review. *Advances in Neural Information Processing Systems*, 2014.